Project Title: A Citizen Science Website Application for Biodiversity Advocacy

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1. Background of Case Study

Environmental monitoring and wildlife conservation have become increasingly important due to rapid urban development, habitat loss, and climate change. In many communities, wildlife sightings are often shared informally through social media or personal communication, which makes it difficult for environmental agencies and researchers to collect accurate and structured data. Without a proper system to record and analyze these observations, valuable environmental information may be lost. Therefore, a digital platform is needed to allow communities to easily record and share wildlife observations in a structured and accessible way.

2. Chosen SDG

This project supports Sustainable Development Goal 13: Climate Action, which focuses on taking urgent action to combat climate change and its impacts. Environmental changes and climate shifts can affect wildlife habitats and biodiversity. By enabling users to record and share wildlife observations, the EcoWatch platform helps collect environmental data that may contribute to better understanding ecological changes and promoting climate awareness within communities.

3. Problem Statement

Many wildlife sightings and environmental observations are not properly recorded or centralized, resulting in a lack of reliable data for monitoring biodiversity and environmental changes. Citizens often encounter wildlife in their surroundings but do not have an accessible platform to report these observations in a structured way. As a result, researchers and environmental authorities have limited data to analyze wildlife distribution and environmental conditions. This creates a gap between public observations and environmental monitoring efforts.

4. Objective

The objectives of the EcoWatch web application are:

1. To develop a user-friendly web application that allows users to submit wildlife observations easily.
2. To collect and store observation data such as species type, location, and images in a centralized cloud database.
3. To provide analytics features that visualize observation data, including number of submissions, species distribution, and location trends.
4. To encourage public participation in environmental monitoring and raise awareness about wildlife conservation.

5. Solution

5.1. A concise Project Description.

The proposed citizen science website functions as a bridge between the public and professional scientific communities by enabling citizens to actively participate in environmental monitoring and data collection. Leveraging the widespread use of smartphones, our website transforms everyday users into “mobile sensors,” significantly expanding the scale and reach of environmental data gathering beyond the limitations of traditional laboratory or agency-based monitoring systems. Through the website, users can record and upload observations of real-world environmental phenomena, such as the variety of plant species, air and water quality. The websites support multimodal input, allowing users to capture photographs, descriptive notes, and other relevant information. Each submission is automatically enriched with contextual metadata, including GPS coordinates and timestamps, ensuring that every observation is geospatially anchored and temporally traceable. The inclusion of visual evidence not only enhances the credibility of the data but also enables researchers and relevant stakeholders to verify findings, such as identifying a specific plant species or locating a potential pollution source.

Beyond data collection, the website allows users to explore environmental observations made by others within their area, helping them recognize patterns, trends, and potential environmental concerns in their immediate surroundings. This shared visibility strengthens community awareness and encourages collaborative participation in environmental stewardship. As users engage with the websites, they develop greater environmental literacy by learning to recognize species, understand ecological indicators, and interpret environmental conditions such as air or water quality. This process increases public scientific awareness while simultaneously empowering communities to participate in environmental advocacy. For instance, communities facing issues such as river pollution or illegal dumping can use the website to collect verifiable evidence and enabling more effective responses. Real-time data submissions also enhance the ability of stakeholders to respond rapidly to emerging environmental

challenges, such as detecting invasive species, identifying biodiversity loss, or monitoring localized ecological disturbances. Furthermore, the aggregated data can serve as an independent source of information for validating the effectiveness of environmental policies, including urban greening initiatives, conservation programs, or biodiversity protection strategies.

In addition to reflecting the use of crowdsourcing, these websites provide two additional functions; i) incorporating AI-powered conversational features (Chatbot) that transform the website from a one-way reporting tool into an interactive and intelligent companion. By offering real-time support and explanations, conversational AI can help reduce common hurdles in citizen science, such as user uncertainty, data entry errors, and lack of scientific knowledge among participants. In addition, ii) implementing gamification elements for user engagement, through leaderboards, that encourage sustained participation and enhance user motivation. These improving long-term user retention and promoting continuous engagement with the citizen science platform. Ultimately, the citizen science mobile website represents more than a simple data collection tool; it forms a socio-technical ecosystem that democratizes scientific participation.

5.2. Technologies Used

The EcoWatch website is developed using **Vite**, a modern front-end build tool that enables fast development and efficient performance for web websites. Vite provides a lightweight development server with instant hot module replacement, allowing developers to see changes in real time while coding. This significantly improves development speed and simplifies the process of building and testing the website interface.

The project is written and managed using **Visual Studio Code (VS Code)**, which serves as the primary code editor and development environment. VS Code provides useful features such as syntax highlighting, debugging tools, integrated terminal access, and extensions that support modern JavaScript frameworks. These features help streamline the development workflow and make it easier for the team to manage the project files, collaborate through version control, and troubleshoot issues during development.

To manage the backend services, the website integrates **Firebase**, a cloud-based platform provided by Google. Firebase is used for several core functionalities within the EcoWatch system. **Firebase Firestore** serves as the real-time cloud database that stores wildlife observation data submitted by users, including information such as species type, location, timestamp, and images. **Firebase Authentication** enables secure user login and account management, ensuring that only authorized users can access certain features of the website. In addition, **Firebase Storage** can be used to store uploaded wildlife images securely in the cloud.

By combining Vite for fast frontend development, Visual Studio Code for efficient coding and project management, and Firebase for scalable backend services, the EcoWatch website achieves a modern and flexible architecture. This technology stack allows the system to support real-time data updates, secure user interactions, and reliable cloud storage, making it well-suited for community-based environmental monitoring and wildlife observation reporting.

5.3. Challenges

Some major challenges involve data quality and reliability, as citizen-generated observations may sometimes contain inaccuracies due to misidentification of species, incomplete information, or inconsistent reporting. Another challenge relates to user participation and long-term engagement**,** since this citizen science websites is relied on the input by crowdsourcing. Maintaining user motivation requires intuitive website design, clear feedback mechanisms, and incentive systems such as digital badges, contribution rankings, or recognition for active contributors. A further challenge concerns digital literacy and accessibility, as some users may have limited experience with mobile websites or may face connectivity constraints, particularly in rural areas. Simplifying the user interface and enabling offline data collection with later synchronization can help address this issue. Additionally, data privacy and ethical considerations must be carefully managed, especially when collecting geolocation data or images that may contain sensitive environmental or community information. Clear consent mechanisms, anonymization practices, and secure data management protocols are therefore essential to protect user trust.

Moreover, other challenges faced was integrating Firebase Storage for storing user-uploaded observations. Initially, the system was designed to allow users to upload images as part of their environmental observations. However, due to service limitation and cost constraints associated with Firebase cloud storage features, the image storage functionality could not be fully implemented. Currently, only the textual user information and observation details are successfully stored in Firebase data storage, while the image files are not saved.

5.4. System Architecture (React + Firebase)

The EcoWatch website follows a modern client–cloud architecture where the frontend interface communicates directly with cloud-based backend services. The system is built using React, a JavaScript library for building interactive user interfaces, while Firebase serves as the backend platform responsible for authentication, data storage, and real-time synchronization.

The frontend layer is developed using React and Vite, allowing the website to provide a responsive and dynamic user experience. React components are used to structure the user interface into modular sections such as the Feed, Observation page, AI Chat interface, Leaderboard, and Analytics dashboard. Each component is responsible for handling specific user interactions and displaying relevant data retrieved from the backend.

Firebase acts as the backend infrastructure that supports several critical services within the website. Firebase Authentication manages user registration and login, ensuring that each user has a unique identity within the system. Cloud Firestore serves as the primary database where all observation data, user information, and activity records are stored. Firestore's real-time data synchronization allows updates made by one user to immediately appear for other users viewing the system. Additionally, Firebase supports secured cloud storage for images uploaded as part of environmental observations.

The overall workflow of the system begins when a user interacts with the website through the React interface. When an observation is submitted, the data—including images, location, timestamp, and descriptive notes—is sent to Firebase Firestore. The database then stores and organizes the information, making it accessible to other website features such as the Feed page and Analytics dashboard. Through this architecture, EcoWatch enables seamless data collection, storage, and visualization while maintaining scalability and reliability.

5.5. Features of EcoWatch

EcoWatch provides several core features designed to encourage community participation in environmental monitoring while also making the website informative and engaging.

Feed

The Feed page serves as the central hub where users can view environmental observations submitted by the community. Each observation typically includes an image, species or environmental description, location information, and the time of submission. This feature allows users to stay informed about environmental activities in their area and observe patterns in biodiversity or environmental conditions.

Observation

The Observation feature enables users to submit new environmental observations directly from their mobile device or browser. Users can upload photos, write descriptions, and allow the system to capture location and timestamp data automatically. This feature forms the core data collection mechanism of the EcoWatch website.

AI Chat

EcoWatch integrates an AI-powered chat assistant that helps users understand environmental topics, species identification, and website-related guidance. The conversational interface allows users to ask questions and receive informative responses in real time, making the website more educational and user-friendly.

Leaderboard

The Leaderboard introduces a gamification element to the website by ranking users based on their contributions. Users who submit more observations or actively participate in the platform can gain recognition through ranking positions. This encourages continuous engagement and motivates users to contribute more environmental data.

Analytics

The Analytics page provides statistical insights derived from the observations collected by the community. Data visualizations such as charts and summaries help users understand trends such as the most frequently observed species, observation counts over time, and user activity levels. These insights transform raw observation data into meaningful information for environmental awareness and research.

5.6. Database Structure

EcoWatch uses Firebase Firestore as its primary database for storing and managing website data. Firestore organizes data into collections and documents, allowing structured and scalable storage of environmental observations and user information.

Users Collection

The users collection stores information related to registered users of the website.

Example fields include:

userId

name

email

points

joinedDate

This data allows the system to track user activity and generate leaderboard rankings.

Observations Collection

The observations collection stores all environmental observations submitted by users.

Example fields include:

species

description

imageUrl

location

timestamp

userId

Each document represents a single observation and contains both user-provided data and automatically captured metadata.

Leaderboard Data

Leaderboard rankings are generated based on user contributions. The website calculates user activity by counting the number of observations submitted by each user. This data is used to determine rankings displayed in the leaderboard feature.

By structuring the database in this way, EcoWatch ensures efficient storage, retrieval, and analysis of environmental data while supporting real-time updates across the website.

5.7. Screenshots Explanation

Screenshots are included in this report to illustrate the main user interface components of the EcoWatch website and demonstrate how users interact with the system.

Login / Authentication Page

This screenshot displays the login interface where users authenticate themselves before accessing the website. Firebase Authentication ensures that only registered users can submit observations or interact with personalized features.

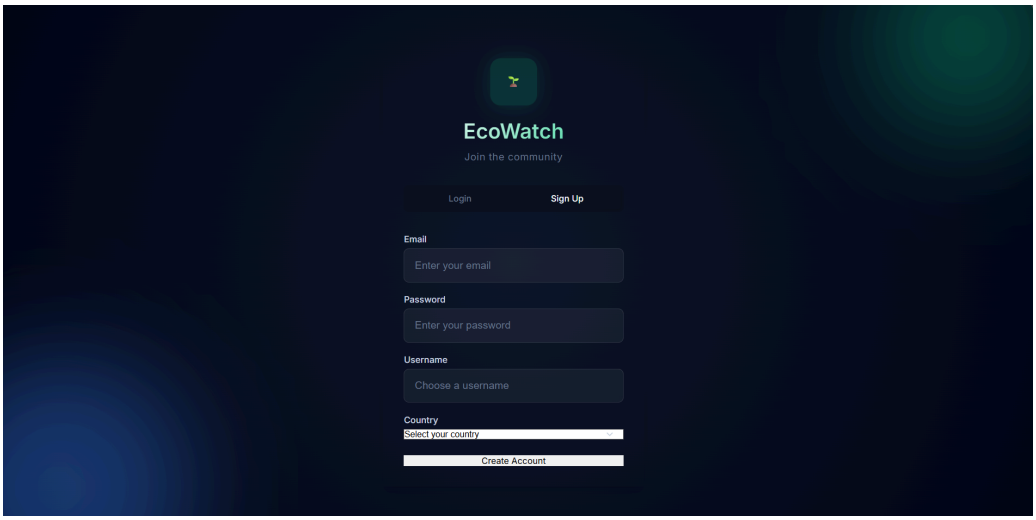

Figure 9.1 : Sign Up page

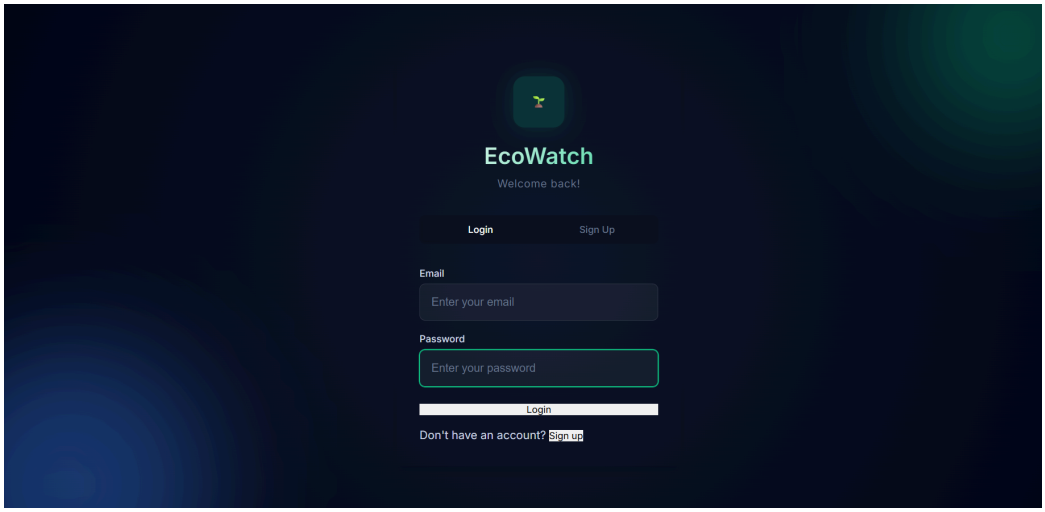

Figure 9.2 : Login page

Feed Page

The Feed page shows a collection of environmental observations submitted by users. Each entry includes details such as an image, description, location, and timestamp. This interface allows users to explore environmental data contributed by the community.

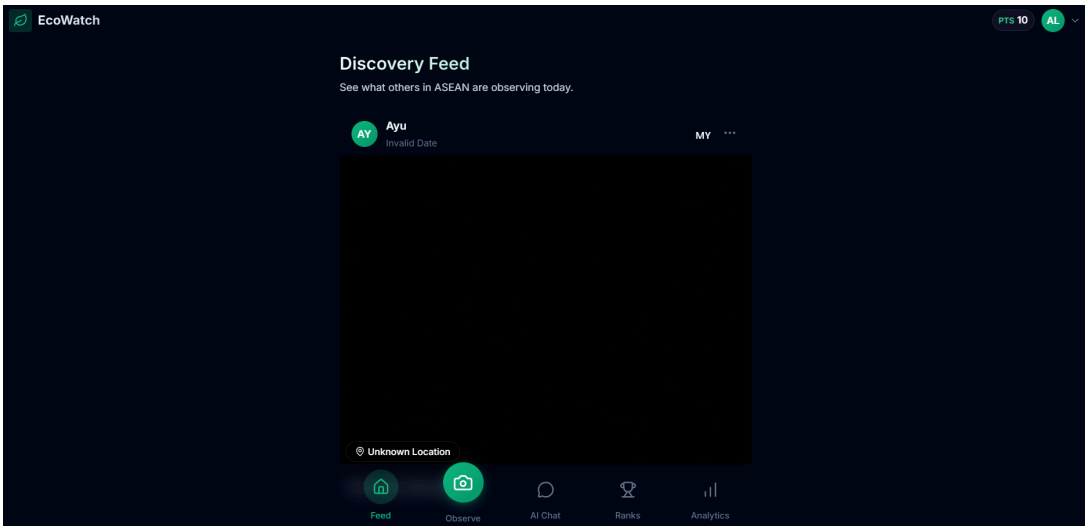

Figure 9.3 Feed Page

Observation Submission Page

This screenshot demonstrates the form used by users to submit new observations. The interface allows users to upload images, write descriptions, and automatically capture location and time data.

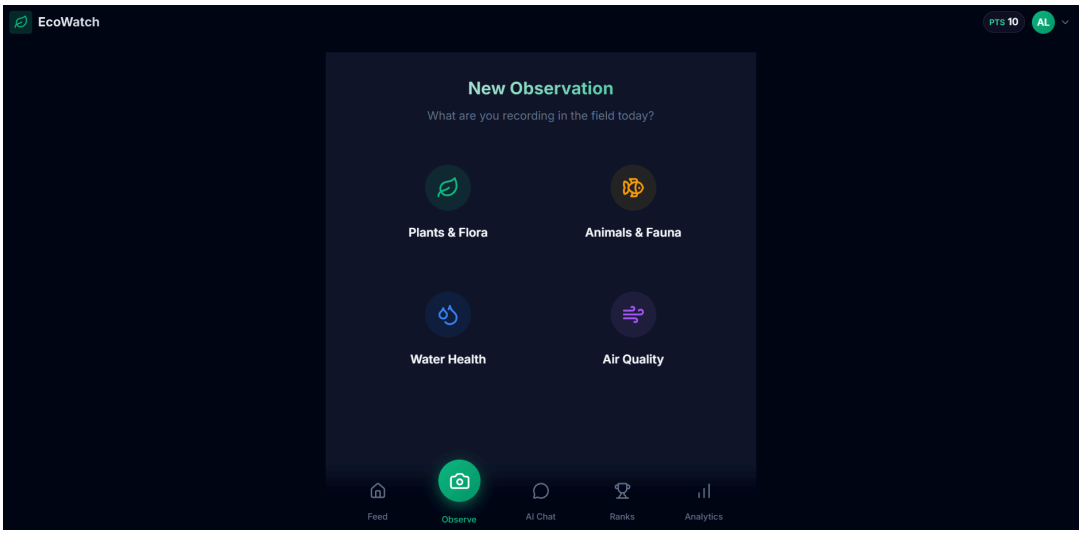

Figure 9.4 Observation Page

Analytics Page

The Analytics page presents visual representations of collected observation data. Charts and summaries help users understand trends in environmental observations and identify patterns within the collected dataset.

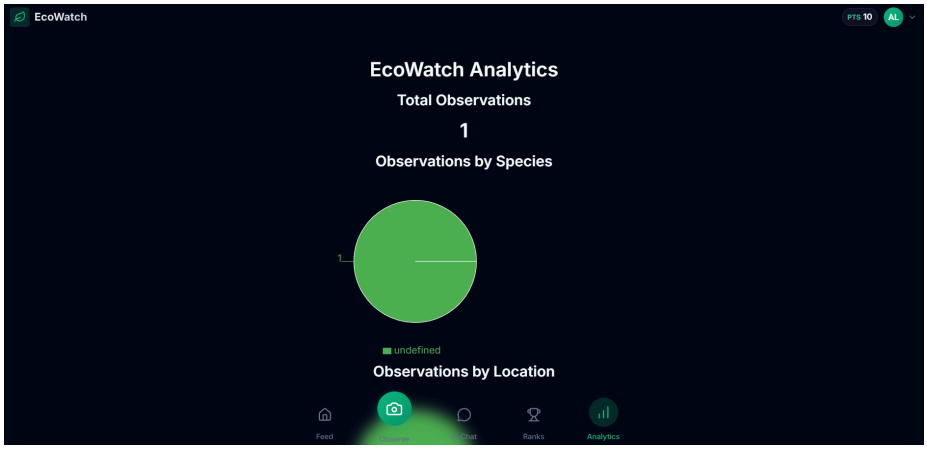

Figure 9.5 Analytics Page

Leaderboard Page

The Leaderboard displays rankings of users based on their contributions to the platform. This feature motivates users to participate more actively in the citizen science initiative.

Figure 9.6 Leaderboard Page

5.8. AI Acknowledgement

This project utilized artificial intelligence (AI) tools to assist in certain aspects of the development and documentation process. The primary AI tool used was **ChatGPT**, which was used to support tasks such as explaining coding concepts, assisting with debugging errors, generating example code snippets, and improving the clarity of written report sections. ChatGPT was also used to help draft and refine descriptions for parts of the report, including the problem statement, objectives, and system explanation.

In addition, **GitHub Copilot** may have been used within the development environment to suggest code completions and assist with writing programming syntax more efficiently during the implementation of the web application.

All AI-generated suggestions were reviewed, edited, and integrated by the project team to ensure that the final implementation and documentation accurately reflect the intended functionality of the EcoWatch system. AI tools were used only as supportive aids to enhance productivity and understanding during the development process.

6. Conclusion

In conclusion, the EcoWatch application was developed to address the lack of structured platforms for recording and monitoring wildlife observations within local communities. By providing an easy-to-use digital platform, the system enables users to submit wildlife sightings, including species information, location, and images. This collected data is stored in a cloud-based database using Firebase, allowing real-time data access and management.

Through the integration of modern development tools such as Vite, Visual Studio Code, and Firebase, the application demonstrates how technology can support community participation in environmental monitoring. The analytics feature also helps visualize observation trends, such as the number of submissions, species distribution, and location data, which can be useful for environmental awareness and research.

Overall, the EcoWatch platform encourages public involvement in wildlife observation and contributes valuable data that can support conservation initiatives. The project also aligns with the objectives of the Sustainable Development Goal 15: Life on Land, which focuses on protecting biodiversity and promoting sustainable ecosystems. In the future, the system can be further improved by adding more advanced analytics, expanding the database, and enhancing user engagement features to support broader environmental monitoring.